

24CST33 - DIGITAL LOGIC AND DESIGN PRINCIPLES

Programme& Branch	B.E. - Computer Science and Engineering	Sem.	Categ ory	L	T	P	SL*	Total	Credit
Prerequisites	Nil	3	ES	45	0	0	45	90	3
Preamble	This course enables the students to understand the basic principles of Binary Codes, Boolean algebra, digital logic gates, combinational and sequential circuits. It also focuses on registers, counters and programmable logic devices								
Unit – I	Number Systems and Boolean Algebra:								9
Complement Number System-Signed Binary Numbers – Binary Codes – Binary Logic - Boolean Algebra: Definitions – Basic and Axiomatic –Theorems of Boolean Algebra – Boolean functions: Realization of functions using Logic gates									
Unit – II	Gate Level Minimization:								9
Canonical and Standard Forms of Boolean functions – Minimization of functions using Karnaugh Map – Don't–Care Conditions – NAND and NOR Implementation– Exclusive-OR function – Minimization of functions using QuineMcCluskey method									
Unit – III	Combinational Logic:								9
Analysis procedure – Design procedure – Binary Adder &Subtractor: Half Adder – Full Adder - Half Subtractor – Full Subtractor– Magnitude Comparator – Decoders – Encoders – Multiplexers – Demultiplexers – Boolean Functions implementation using Multiplexers									
Unit – IV	Sequential Logic:								9
Introduction – Latches and Flip-flops – Analysis of clocked sequential circuits: State Equations – State Table – State Diagram – State Reduction and Assignment– Mealy and Moore machines and their circuit design procedure.									
Unit – V	Register, Counter								9
Register, Counter : Shift Registers: Serial Transfer – Serial Addition – Universal Shift register – Binary Ripple Counter – BCD Ripple Counter – Ring Counter – Johnson Counter.									
TEXT BOOK:									
1.	Morris Mano M., Micheal D. Ciletti, "Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog", 6th Edition, Pearson Education, 2024.								
REFERENCES/ MANUAL / SOFTWARE:									
1.	Salivahanan S. &Arivazhagan S., "Digital Circuits and Design", 5th Edition, Oxford University Press, New Delhi, 2022.								
2.	Donald D.Givone , "Digital Principles and Design", Tata McGraw-Hill 1st Edition,2012.								
3.	Virtual Labs: http://vlabs.iitkgp.ac.in/dec								

*includes Term Work(TW) & Online / Certification course hours

COURSE OUTCOMES: On completion of the course, the students will be able to		BT Mapped (Highest Level)
CO1	apply the different number systems and their conversion and boolean algebra	Applying (K3)
CO2	make use of map and tabulation technique to evaluate the given boolean expression	Applying (K3)
CO3	make use of combinational logic circuits to evaluate the boolean expression	Applying (K3)
CO4	apply the concepts of sequential logic circuits to implement boolean functions	Applying (K3)
CO5	construct simple digital systems using registers and counters.	Applying (K3)

Mapping of COs with POs and PSOs

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1									3	1
CO2	3	2	1									3	1
CO3	3	2	2	2	1					1		3	1
CO4	3	2	2	2	1					1		3	1
CO5	3	2	2	2	1					1		3	1

1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy

ASSESSMENT PATTERN – THEORY

Test / Bloom's Category*	Remembering (K1) %	Understanding (K2) %	Applying (K3) %	Analyzing (K4) %	Evaluating (K5) %	Creating (K6) %	Total %
CAT 1		45	55				100
CAT 2		50	50				100
CAT 3		50	50				100
ESE		40	60				100

* ±3% may be varied (CAT 1,2,3 – 50 marks & ESE – 100 marks)


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 Board of Studies - CSE



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 [T. Kumaravel]

24MNT31 - ENVIRONMENTAL SCIENCE									
(Common to All Engineering and Technology Branches)									
Programme & Branch	All B.E/B.Tech Branches	Sem.	Category	L	T	P	SL*	Total	Credit
Prerequisites	NIL	3 / 6	MC	30	0	0	0	30	0
Preamble	This course provides an approach to understand the various natural resources, ecosystem, bio-diversity, pollution control & monitoring methods for sustainable life and also to provide knowledge and to create awareness for engineering students onsocial Issues and the environment.								
Unit – I	Environmental Studies and Natural Resources							6	
Introduction to Environmental Science – uses, over-exploitation and conservation of forest, water, mineral, food, energy and land resources–case studies.									
Unit – II	Ecosystem and Biodiversity							6	
Ecosystems: concept and components of an ecosystem -structural and functional features – Functional attributes (Food chain and Food web only). Biodiversity: Introduction – Classification – Bio geographical classification of India- Values of biodiversity – Threats and Conservation of biodiversity - case studies.									
Unit – III	Environmental Pollution							6	
Environmental Pollution: Definition – causes, effects and control measures of:(a) Air pollution - Climate change, global warming, acid rain, ozone layer depletion (b)Water pollution (c) Soil pollution - Role of an individual in prevention of pollution - case studies.									
Unit – IV	Environment Quality Standards and Monitoring							6	
Ambient air quality standards - Water quality parameters and standards: Turbidity, pH, Suspended solids, hardness, residual chlorine, sulfates, phosphates, iron and manganese, DO, BOD, COD (definition, specifications and limits only) - Introduction to EIA – objectives and process of EIA - environment protection act – air (prevention and control of pollution) act – water (prevention and control of pollution) act- case studies									
Unit – V	Social Issues and the Environment							6	
From Unsustainable to Sustainable development - three pillars of sustainability- factors affecting environmental sustainability- approaches for sustainable development- Social issues: Urban problem related to energy - population growth and explosion - issues related to resettlement and rehabilitation, E-waste recycling - role of IT in environment and human health - case studies.									
TEXT BOOK:									
1.	AnubhaKaushik, and Kaushik C.P., “Environmental Science and Engineering”, 6th Multicolour Edition, New Age International Pvt. Ltd., New Delhi, 2023.								
REFERENCES:									
1.	Palanisamy P.N., Manikandan P., Geetha A., Manjula Rani K., Kowshalya V.N., “Chemistry for Engineering”, Revised Edition ,Pearson Education, New Delhi, 2024.								
2.	ErachBharucha, —Textbook of Environmental Studies for Undergraduate Coursesll, University Grands Commission, Universities Press India Private Limited, Hyderguda, Hyderabad, 2005.								

*includes Term Work(TW) & Online / Certification course hours

COURSE OUTCOMES: On completion of the course, the students will be able to		BT Mapped (Highest Level)
CO1	illustrate the various natural resources and role of individual for its conservation	Understanding (K2)
CO2	elaborate the features of ecosystem and biodiversity to find the need for conservation.	Understanding (K2)
CO3	manipulate the sources, effects and control methods of various environmental pollution.	Applying (K3)
CO4	make use of the knowledge of Quality standards, EIA and environmental legislation laws to monitor the environment.	Applying (K3)
CO5	utilize the knowledge of various social issues and impact of population explosion on environment towards sustainability.	Understanding (K2)

Mapping of COs with POs and PSOs

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	1	3			2	1						
CO2	2	1	3			2	1						
CO3	2	2	3			2	1						
CO4	2	2	3			2	1						
CO5	2	1	3			2	1						

1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy

ASSESSMENT PATTERN - THEORY

Test / Bloom's Category*	Remembering (K1) %	Understanding (K2) %	Applying (K3) %	Analyzing (K4) %	Evaluating (K5) %	Creating (K6) %	Total %
CAT1	25	40	35				100
CAT2	25	40	35				100
CAT3	25	40	35				100
ESE	NA						

* ±3% may be varied (CAT 1, 2 & 3 – 50 marks)

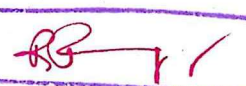

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 Board of Studies - *Sarth*



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24CSL31 -DATA STRUCTURES LABORATORY													
Programme& Branch		B.E. – Computer Science and Engineering				Sem.	Category	L	T	P	SL*	Total	Credit
Prerequisites		Programming and Linear Data Structures				3	PC	0	0	30	0	30	1
Preamble		This course provides knowledge to develop applications using the concepts of linear and non-linear data structures.											
LIST OF EXPERIMENTS / EXERCISES:													
1.	Implementation of Linked List and its Operations.												
2.	Implementation of Polynomial Addition and Balancing Parenthesis using ADT.												
3.	Implementation of Binary Search Tree Traversals.												
4.	Implementation of AVL Tree Operations.												
5.	Implementation of Heap and its Operations.												
6.	Implementation of Graph Traversal Techniques.												
7.	Implementation of Topological Sorting Algorithm.												
8.	Implementation of Dijkstra's Algorithm.												
9.	Implementation of Prim's and Kruskal's Algorithm.												
10.	Implementation of Hash Table Operations using an Array: Store Elements, Search Elements, and Delete Elements.												
11.	Mini Project												
REFERENCES/ MANUAL /SOFTWARE:													
1.	Operating System : Windows/Linux												
2.	Software : C/Java												
3.	Laboratory Manual												
COURSE OUTCOMES: On completion of the course, the students will be able to												BT Mapped (Highest Level)	
CO1	utilize linear data structures effectively to address various computational problems.											Applying (K3), Precision (S3)	
CO2	apply concepts of linear data structures to solve problems typically associated with non-linear data structures.											Applying (K3), Precision (S3)	
CO3	select appropriate graph algorithms and indexing techniques to optimize the performance of algorithms.											Applying (K3), Precision (S3)	
Mapping of Cos with POs and PSOs													
COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	2	1							3	2
CO2	3	2	1	2	1							3	2
CO3	3	2	1	2	1							3	2
1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy													

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V. MANIMARAN



24CSL32 - DESIGN THINKING LABORATORY

Programme & Branch	B.E. – Computer Science and Engineering	Sem.	Category	L	T	P	SL*	Total	Credit
Prerequisites	NIL	3	PC	0	0	30	0	30	1

Preamble It provides an exposure to develop a prototype model for design challenge.

LIST OF EXPERIMENTS / EXERCISES:

1.	Develop SCOPES Template for your design challenge.
2.	Perform User Research by using explore method and tools.
3.	Conduct Field Observations for your design challenge.
4.	Conduct an interview with your customer by using empathy map and journey map.
5.	Create user personas for your product or service.
6.	Develop SCAMPER template for ideation.
7.	Create user scenarios / story telling for your product or service.
8.	Create low-fidelity prototypes (paper prototypes) for your design challenge.
9.	Create medium fidelity prototypes (hardware/software prototypes) for your design challenge.
10.	Collect feedback from users for your prototype model.
11.	Mini Project

REFERENCES/ MANUAL /SOFTWARE:

1.	Chart papers, Sticky Notes, IOT Components.
2.	Lee Chong Hwa, "Design Thinking the Guidebook", Design Thinking Master Trainers of Bhutan, 2017. (E-book)

COURSE OUTCOMES:

On completion of the course, the students will be able to

**BT Mapped
(Highest Level)**

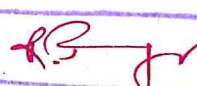
CO1	construct design challenge and reframe the design challenge to design opportunity.	Applying (K3), Precision (S3)
CO2	interview the user, and know the feelings of users to foster deep understanding and be able to uncover the deep user insights and needs.	Applying (K3), Precision (S3)
CO3	develop ideas and prototypes by brain storming using the ideation tools.	Applying (K3), Precision (S3)

Mapping of Cos with POs and PSOs

COs/POs/PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	1				3	2	2	3	1
CO2	3	3	3	2	1				3	2	2	3	1
CO3	3	3	3	2	1				3	2	2	3	1

1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy

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